

Jason Blackert

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- Embedded Software Engineer with 6+ years in firmware, RF systems, and IoT across startup and enterprise environments specializing in real-time systems like STM32.
- Delivered productionized solutions including TCP-based bootloaders, adaptive thermal controls, and polynomial-based battery estimation models for embedded platforms.
- Improved RF transmit performance by 6dB (low-band) and 3dB (high-band) through hardware-software co-optimization, improving signal strength in mission critical applications.
- Strong background in embedded C/C++, Python, and protocols like Modbus, TCP, MQTT, CAN, and SPI.
- Experienced in environmental and energy-oriented IoT systems, including gateway development, sensor fusion, and edge processing for solar and mobile platforms.

PROFESSIONAL EXPERIENCE

CACI International Inc.

July 2024 – Present

Embedded Software Engineer (Technical Level 4) – December 2025 - Present

- Systematically tested, identified and resolved firmware bugs in software steerable array that was hindering RF performance - increased power output by ~30dB and vetted software suite providing reliable control.
- Redesigned the transmit calibration enabling higher transmit power by pushing into compression and avoiding saturation; giving the highest drive level without distributing energy into different bands.

Embedded Software Engineer (Technical Level 3) – January 2025 - November 2025

- Improved RF transmission performance by 6dB in low bands and 3dB in high bands through tuning and hardware-software optimization, enhancing signal quality in diverse environments.
- Created a polynomial-based battery estimation algorithm with ~5% accuracy to overcome hardware limitations, demonstrating strong applied math and modeling skills in realized systems.
- Led comprehensive benchtop, electrical, RF, firmware, and calibration tests, ensuring system-level functionality and compliance for production-grade embedded systems.

Embedded Software Engineer (Technical Level 2) – July 2024 - January 2025

- Developed a TCP-based bootloader enabling over-the-air firmware upgrades, reducing unit servicing time and increasing functional development speed by 30–40%. Eliminated need to open hardware, preserving RF chain integrity during testing.
- Collaborated cross-functionally with mechanical engineering to design and implement an adaptive thermal management system. Developed a sensor-specific fan curve algorithm that reduced operating temperature by 20% in extreme conditions.

Escamilla & Poneck LLP

September 2023 – July 2024

Senior Embedded Software Engineer – September 2023 - Present

- Spearheaded the digitalization of confidential client data onto encrypted Linux servers utilizing C/C++ and on embedded systems, reducing office paperwork by 90%.
- Architected web framework for seamless data access, improving remote accessibility and cutting response time by 30%.
- Introduced local access via mounted network drive, improving retrieval speed by 25% saving on average 15 hours per week for employees.

Yotta Energy Co.

October 2022 – August 2023

IoT Software Engineer – October 2022 – August 2023

- Engineered an innovative IoT gateway supporting MQTT communication for solar energy storage systems, increasing effectiveness by 25% through tokenized acknowledgements in Python and embedded C.

- Created Yocto recipes for custom distribution, wrote device drivers and deployed docker-packaged services.
- Successfully initialized M.2 cellular modem connection, reduced dropouts by 90%, and ensured uninterrupted connectivity for critical operations.
- Implemented Modbus RTU to enhance accuracy in reading revenue-grade power meter data. Expanded functionality with Modbus TCP integration for modeling software, enabling peak shaving, time-of-use arbitrage, and demand-discharge, resulting in a 30% performance improvement.

Tactical Labs Co.

June 2019 – September 2022

Design Engineer – January 2020 – September 2022

- Leveraged USART and CAN communication for real-time 2-D visualization of femtofarad-level RF sensing technology, facilitating hardware analysis, and improving decision-making process.
- Developed bootloader to validate FOTA packages, safeguarding firmware updates with backup protection to ensure 100% reliability, thus minimizing system downtime.
- Ported sensing classifiers from Python to embedded C and Python to C++, and improved confidence rate from 95.0% to 99.9% in a high precision sensing system.
- Developed and brought up firmware for custom PCB and made modifications on redesign(s).

Electrical Engineer – August 2019 – December 2019

- Integrated PyQt5 application with regression tester for image analysis, confirmed high repeatability in sensing algorithms on a real-time operating system (RTOS).
- Adapted Unity Engine for CES Demo, highlighting gesture classification engine for Automotive applications, fostered engagement with automatic autopilot simulation, enhancing user understanding and product visibility.

Electrical Engineer Intern – June 2019 – August 2019

- Created automated regression testing solution with RoboDK, streamlining testing processes and improving efficiency by 35%.

EDUCATION

Texas State University

August 2015 – December 2019

Bachelor of Science in Electrical Engineering, Computer Engineering

Undergraduate Researcher – January 2018 – August 2019

- Enhanced sound localization techniques utilizing impulse localization in an FPGA environment and designed convolution neural network models to accelerate processing time by 5%; analysis presented in MATLAB.

Capstone Project – January 2018 – August 2019

- Drone Battery Management System – leveraged SPI to connect NXP's S32K to their development board to monitor battery and take action accordingly.

TECHNOLOGY & TOOLS

Languages:

Python | Embedded C | C++ | C# | Verilog | VHDL | JavaScript

Databases:

MongoDB | InfluxDB | RedisDB | MySQL | NoSQL

Frameworks:

Agile | Jira | Debian | AWS IoT | Ansible | Poetry | Pandas | Matplotlib | Django | OpenCV | TensorFlow | PyQt | Qt | RoboDK

Communication Protocols:

MQTT | GSM/LTE | BLE | TCP | UDP | UART/USART | CAN bus | Modbus RTU/TCP (RS-485) | I2C | GPIO | SPI | LAN

Chipsets:

CM4 | SimCom 7600 | ARM M7 | Atmel V71 | Atmel E51 | STM32 F, H, K, and G-series